

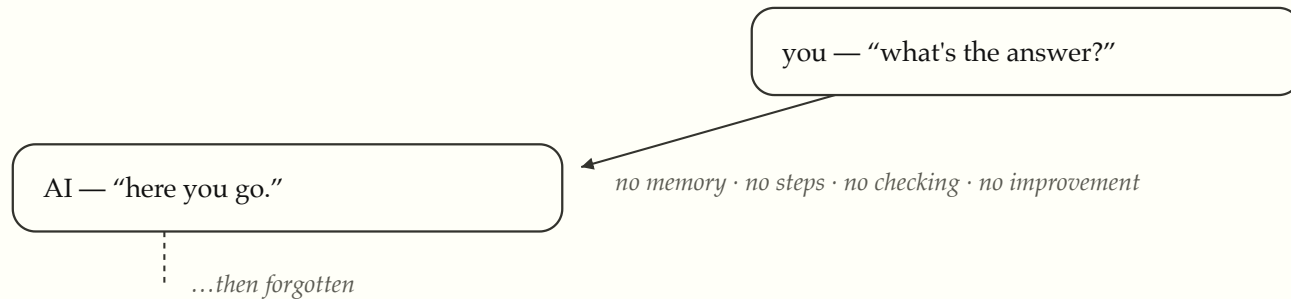
Intro to AI Engineering

The craft of building reliable systems around AI — not just better prompts.

BEFORE WE START, A WARNING

The trap: AI as an answer machine.

Most people meet AI through a *chatbot* — type a question, get an answer. Handy, but it quietly trains you to treat AI as a vending machine for facts.



One question in, one answer out — then forgotten. Useful, but it's only the bottom rung of the ladder.

The rest of this deck is about everything that view leaves out.

FIRST, THE BIG PICTURE

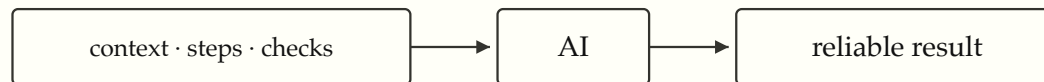
What is AI engineering?

The craft of *designing the setup* around an AI model so it reliably does useful work — not fishing for the one clever prompt that happens to land.

just asking



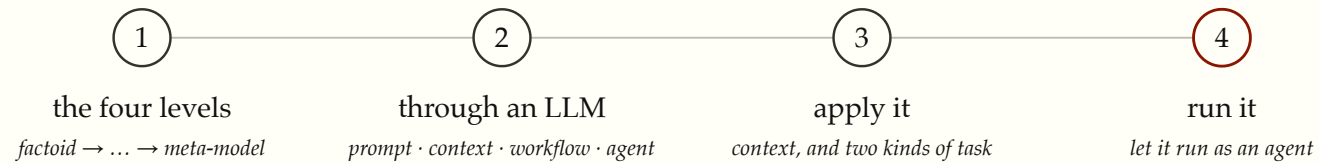
AI engineering



Same model. The difference is the structure you build around it.

THE PLAN

Where we're headed.

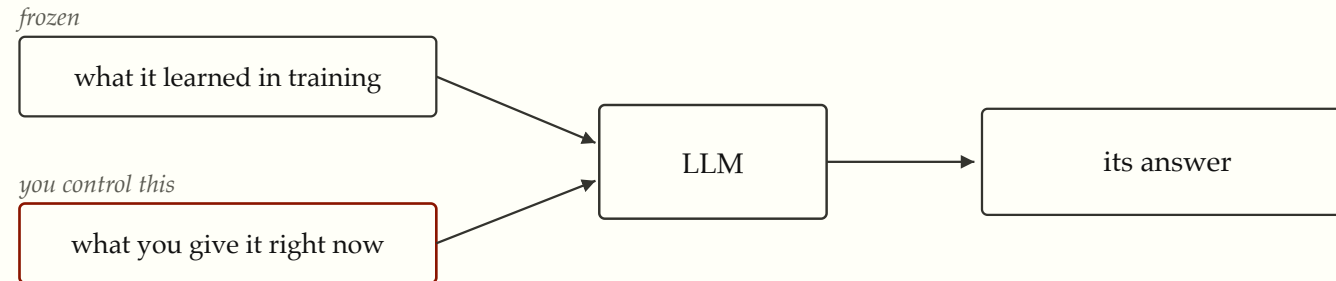


Four levels, how they land on an LLM, how to apply them — ending with a real agent you can run.

WHY IT'S EVEN NEEDED

An LLM is powerful, but blind.

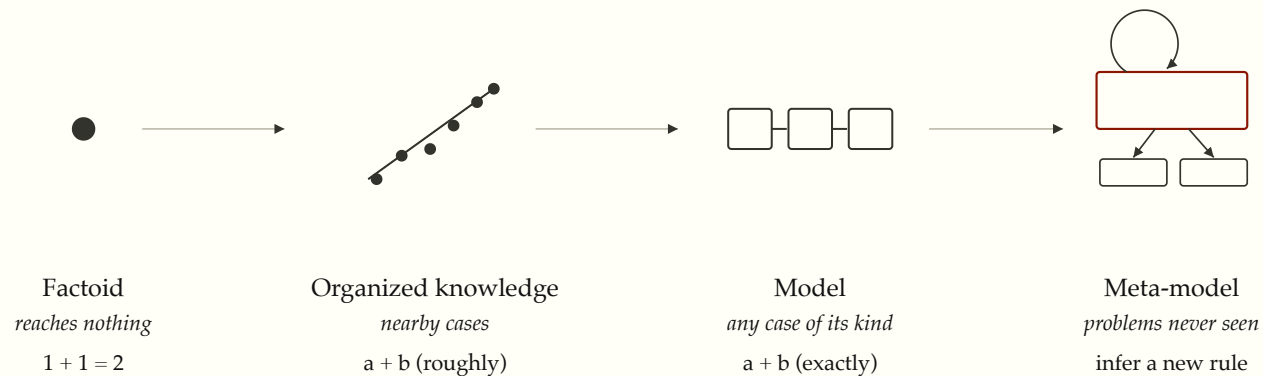
It knows only two things: the patterns it soaked up in training, and whatever you put in front of it right now. Give it nothing else and it will guess — confidently.



Left alone, it fills the gaps by guessing. Structure — context, clear asks, steps, limits — turns raw power into reliable work. That's the ladder you climb next.

Each level is made of the one below it.

Start with one stored answer. Find the rule across many. Turn the rule into an exact model. Then build something that makes new models on its own — each step abstracts further and reaches problems the one below it could not.

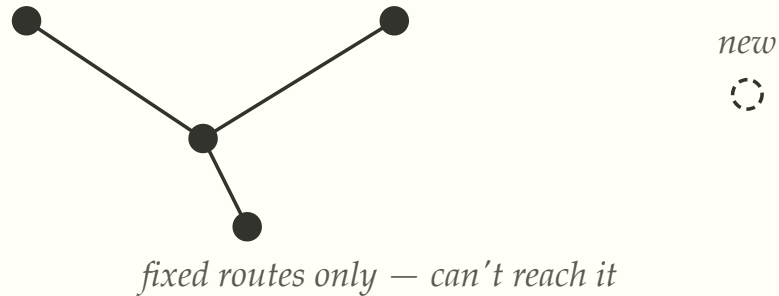


Move right and you climb in abstraction — never leaving the level below behind.

Two kinds of intelligence.

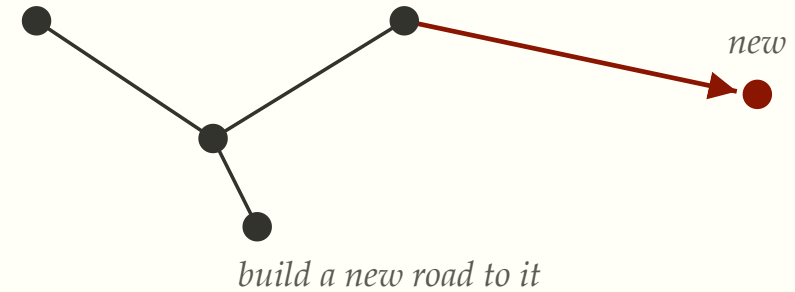
CRYSTALLIZED SKILL

A library of procedures for problems you've already met. Fast and reliable — until the problem is genuinely new. A *road network*: fixed routes between fixed places.



FLUID INTELLIGENCE

Building a new procedure for a problem you've never seen — from just a few examples. A *road-building company*: lay a road wherever you need to go.



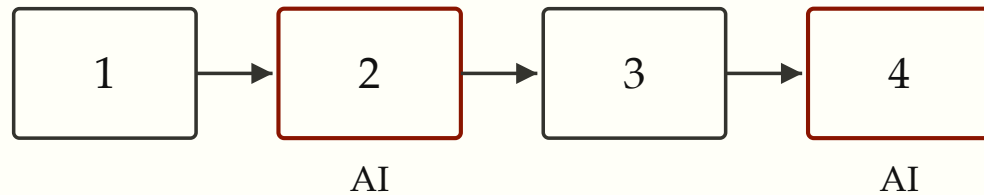
Skill is not intelligence. On the spectrum, the first three rungs are skill — only the top, the **meta-model**, is the fluid kind.

THE SAME SPLIT, IN YOUR WORK

Exploit vs. explore.

EXPLOIT WHAT YOU KNOW

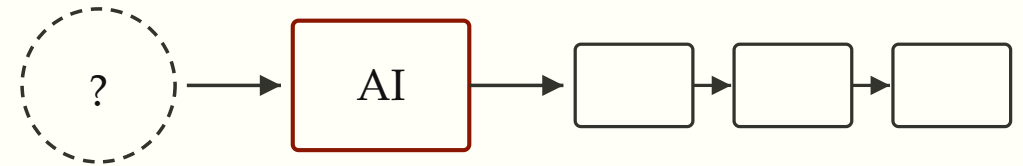
There's a known procedure. Map it into steps and hand the routine ones to AI — *crystallized skill* is enough.



map it, then apply AI to some steps

EXPLORE THE UNKNOWN

There's no procedure to follow. You and the AI have to *invent* one — this is where the *fluid* kind earns its keep.



AI helps invent a process

Match the task to the kind of intelligence it needs — and most of the value is in the second.

PHASE 1

Factoid

A STORED ANSWER, NO VARIABLE.

The bottom of the spectrum: a single memorized point. $1 + 1 = 2$, kept as a bare answer, tells you nothing about $2 + 3$. It *generalizes to nothing* — store a million and you're still stuck the moment a question doesn't match one you kept.

Degree zero of abstraction — pure memory, no reach.



a factoid

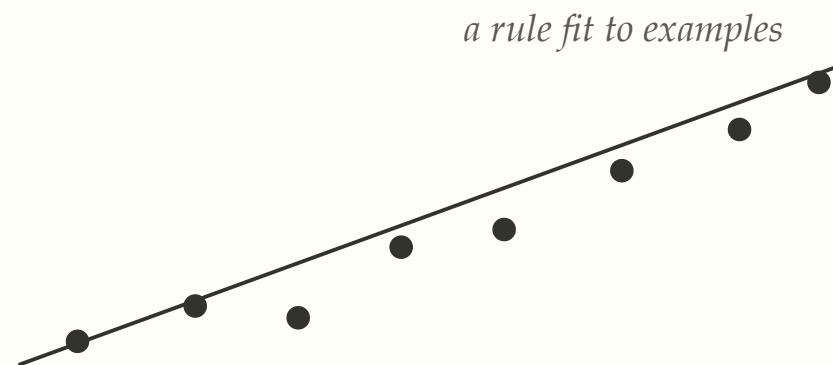
PHASE 2

Organized knowledge

A RULE FIT TO EXAMPLES.

Gather many factoids and a pattern appears: you learn *addition* as a rule with a variable, $a + b$. It reaches beyond what you saw — but only roughly. Solid on familiar sums, shaky on huge or unusual ones. It's how an LLM "adds," and why it slips.

Weak abstraction — interpolation between cases you've seen.



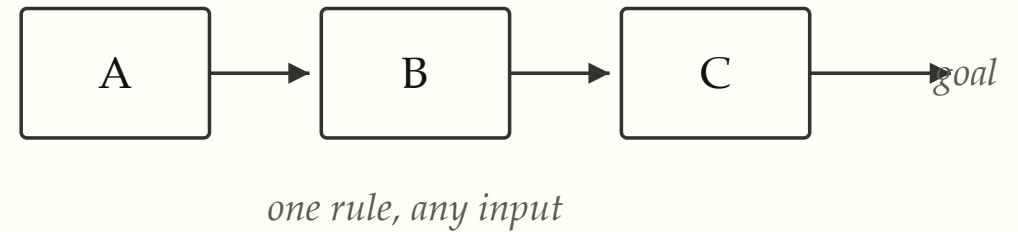
PHASE 3

Model

AN EXACT, CAUSAL RULE.

A model captures how the answer is actually produced — not a pattern guessed from examples. *Addition* as an algorithm: line up the digits, carry the 1. It's right for *any* two numbers, however large, seen before or not — correct 100% of the time.

Strong abstraction — concise, causal, exact.



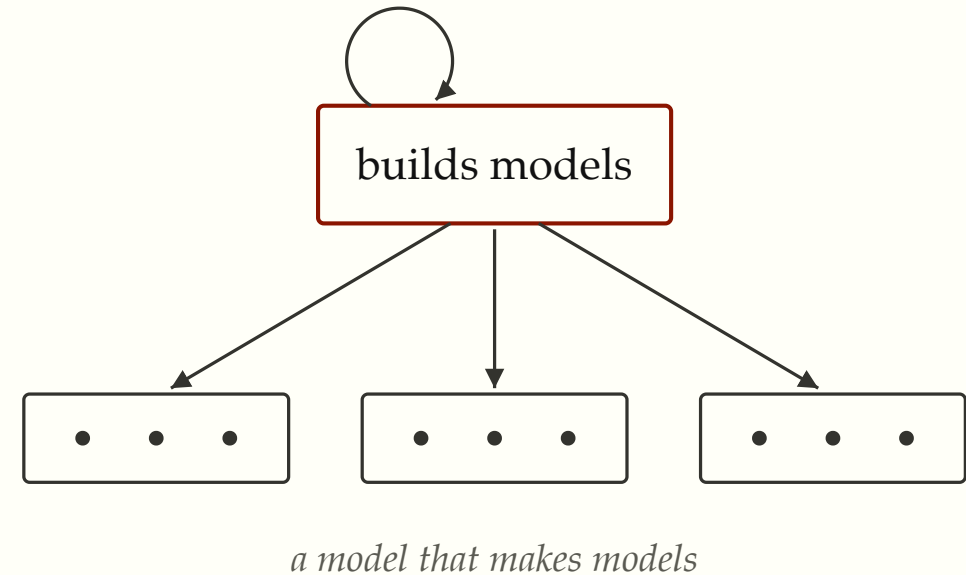
PHASE 4

Meta-model

IT BUILDS NEW MODELS.

The top of the spectrum doesn't apply a model — it *makes* one for a problem it was never prepared for. Show it two examples of a brand-new operation and it infers the rule itself, from almost nothing. A road network gets you between fixed points; this is the *road-building company*.

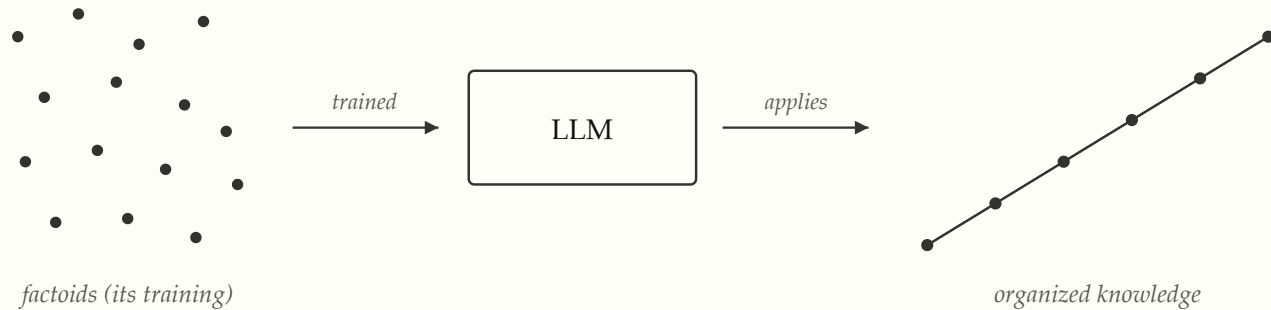
Generating abstractions on the fly — this is intelligence.



NOW BRING IN THE LLM

What is an LLM, in these terms?

It was trained on oceans of *factoids* and squeezed them into *organized knowledge* — patterns it applies to things it never saw.



An LLM lives low on the ladder — strong on factoids and organized knowledge, but short of an exact model. The higher rungs you build around it.

THE FOUR LEVELS, THROUGH AN LLM

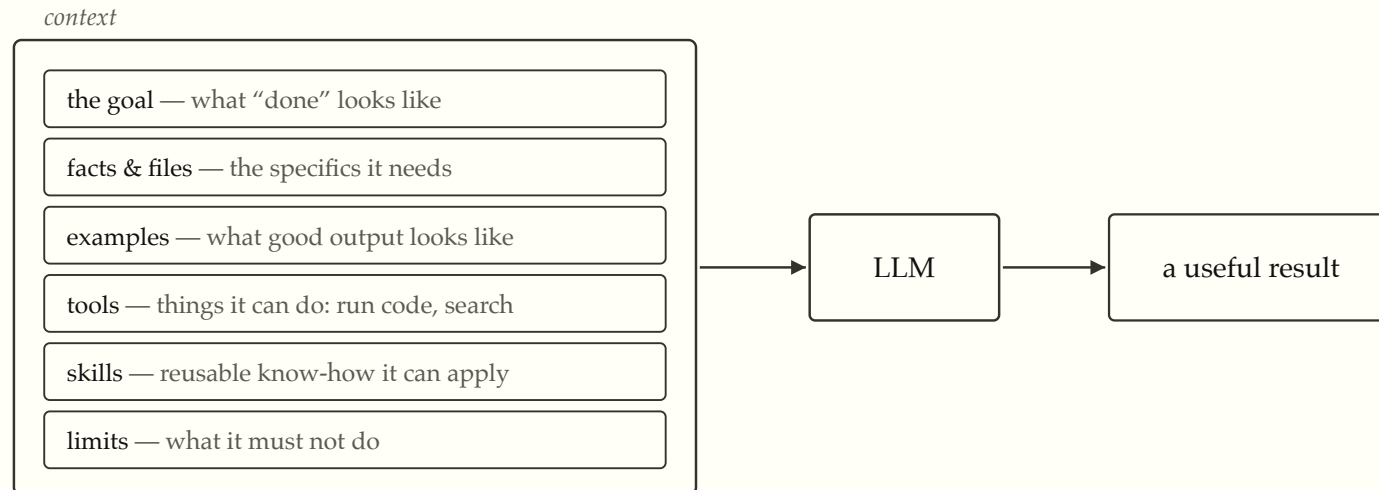
You climb by what you add around it.

- FACTOID → write a prompt — a single, direct ask
 - ORGANIZED KNOWLEDGE → assemble its context — the organized body of facts, files, and examples it draws on
 - MODEL → wire a workflow — a fixed, reliable procedure
 - META-MODEL → let it run as an agent — it builds its own procedure
-

Same model every time — the rung changes with how much structure you wrap around it.

It's all about context.

Context is everything you put in front of the model so it can do the job — its goal, the facts and files that matter, examples of good work, the *tools* it can use, the *skills* it can apply, and the limits.



Tools and skills are part of context too — they're what turn a chat model into something that can act. Better context, better results.

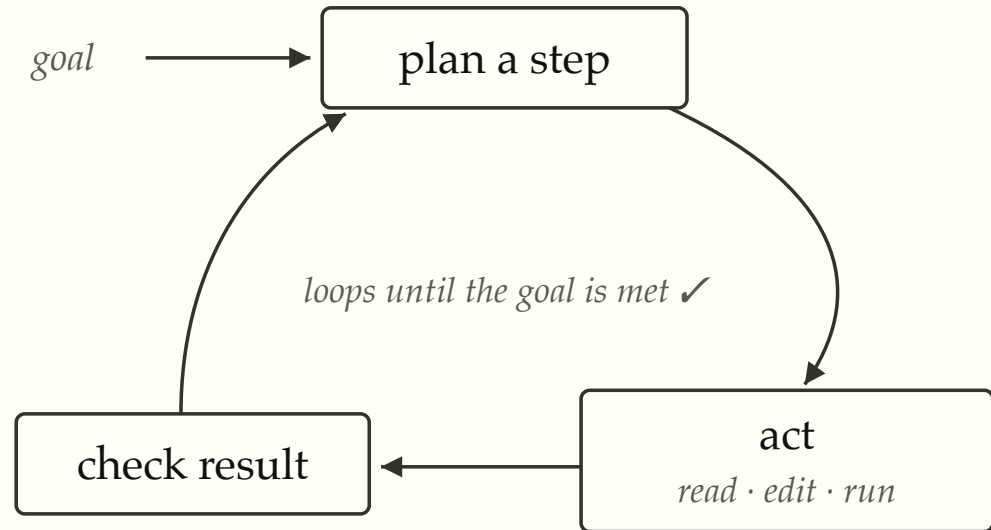
AI engineering is mostly the craft of building the best context.

THE TOP RUNG, UP CLOSE

What "agent" means

An agent runs an *agent loop*. Hand the LLM a goal and some tools; it plans a step, takes it — reads a file, edits code, runs a command — looks at what happened, and loops until it's done.

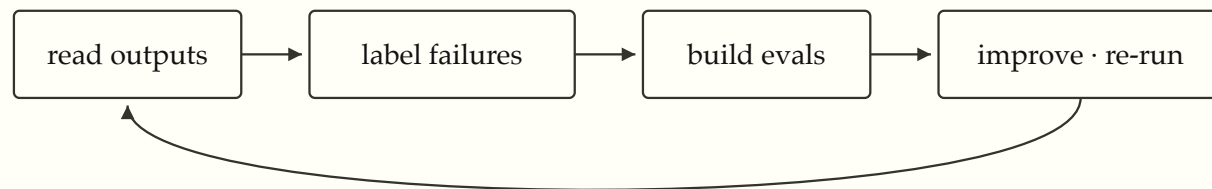
A meta-model at work: it writes its own procedures, then runs them.



HOW DO YOU KNOW IT'S WORKING?

Evaluation.

Not a vibe, and not a generic leaderboard. *Read your own outputs*, name the specific ways they fail, turn those failure modes into repeatable tests — then fix and re-run.



your real failures define your tests — not a leaderboard

| *Look at your data.*

TO WRAP UP

From answer machine to engineer.

You no longer just ask for answers — you build the setup the model works inside: **prompt** → **context** → **workflow** → **agent**, and you judge what comes back.

Whatever the rung, three things stay yours: **the goal, the standards,** and **the limits.**

That's the shift: not a better answer, but a system you can trust.

FOR REFERENCE

Glossary.

AI machines doing things that used to need human thinking.

MODEL the trained network itself; an LLM is one kind.

CONTEXT everything you put in front of the model to do the job.

TOOL an action the model can call: search, run code, edit a file.

WORKFLOW a fixed chain of steps run in order.

COMMAND a single instruction you run in the terminal.

HALLUCINATION a confident answer that's simply wrong.

LLM large language model — the text-prediction engine behind today's AI.

PROMPT the instruction you give the model for one task.

TOKEN the chunk of text a model reads and writes ($\approx$ a word-piece).

SKILL a reusable rule or know-how the model applies.

AGENT a model that plans its own steps and loops until a goal is met.

EVALUATION a test that scores whether outputs are good (an "eval").

OPENCODE the open-source terminal tool we use to build agents.